

# SATYAM PATIL

Maharashtra, India

📞 +91 9370162911 ✉ satyam.content@gmail.com 🔗 LinkedIn 🌟 GitHub

## EDUCATION

### Indian Institute of Technology, Mandi

B.Tech, Computer Science and Engineering (CGPA of 8.25)

Nov 2021 – Jun 2025

Himachal Pradesh, India

### TU Dresden

Semester Exchange Student in Computer Science and Engineering (SGPA of 9)

Oct 2023 – Feb 2024

Dresden, Germany

## TECHNICAL SKILLS

**Languages:** Python, C++, TypeScript, JavaScript, SQL, Kotlin, Dart

**AI Engineering:** LLM agents (ReAct loops, tool calling, multi-agent orchestration), RAG pipelines, FAISS, FastEmbed, LangGraph, MCP, knowledge graphs (Neo4j, Cypher), OpenAI / Anthropic / Gemini APIs

**Backend & Data:** FastAPI, PostgreSQL, async Python, Flask, NodeJS, ReactJS, VueJS, React-Native, Pandas, NumPy, scikit-learn, Git, Docker

**Systems:** Android / AOSP (Soong, Android.bp), SEAndroid & SELinux policy, TEE–Rich OS interfaces, Linux

## EXPERIENCE

### Samsung Research

Research Engineer | LLM Agents, RAG, Android Security, Automation

June 2025 – Present

Noida

- Solved automation on a codebase no external LLM can touch: built **AX\_Agent**, an in-house **ReAct-loop agentic system** running entirely on internal infrastructure, exposing security and porting scripts as agent tools so engineers can drive multi-step porting and policy tasks in natural language.
- Solved recurring-issue knowledge loss: built **CoreBrain**, a **RAG system** over internal JIRA and project-lifecycle issues plus their resolution commits, summarizing each into a problem/solution pair so a newly filed issue retrieves its closest prior fix instead of being re-diagnosed from scratch.
- Solved policy-misconfiguration triage at scale: modeled SELinux policy rules as a graph, trained a **Graph Neural Network** to learn relationships between allow-rule pairs, then scored each node with a **Variational Autoencoder's** reconstruction error to separate well-formed policy from misconfigured entries without hand-written rules.
- Diagnosed and resolved **100+ SEAndroid and security module issues** spanning policy misconfigurations, access violations, and system-call denials across the TEE↔Rich OS boundary; owned the **ICCC module** end-to-end across both the trusted app and its Android-side client service.

### Brainwave Science

Full Stack Developer Intern | VueJS, PostgreSQL, Fastify, PsychoPy, React-Native

Feb 2024 – Dec 2024

Remote

- Unified three previously separate systems onto one **full-stack platform**: browser-based PsychoPy experiments (normally desktop-only), low-latency RTSP IP-camera streaming, and a cross-platform educational app serving **500+ users**.

### LASR Lab, TU Dresden

Application Development Intern | Kotlin, Gradle, Machine Learning, Computer Vision

Oct 2023 – Jan 2024

Dresden, Germany

- Solved real-time object detection under embedded constraints: built an **Android application** for the Digit tactile sensor with **on-device inference**, cutting false detection rates by **30%**.

## PROJECTS

**Rankit – AI Mock-Test Platform (Co-founder)** | FastAPI, PostgreSQL, LangGraph, FAISS, OpenAI API | [rankit.in](https://rankit.in)

- Operate a **live JEE mock-test platform** end-to-end for real students (FastAPI, PostgreSQL, Vite frontend, and the AI analysis layer), deployed across Vercel, Fly.io, and Neon.
- Solved root-cause weak-topic detection instead of surface-level tagging: built a **multi-agent analysis backend** where post-test SQL aggregates feed an LLM with query routing, task decomposition, and tool selection, backed by a **knowledge-graph layer** (topics as nodes, prerequisites as edges) that traces each gap back to its prerequisite instead of flat topic similarity.
- Solved reliable classification of scanned exam questions, including embedded Word equations, into a deduplicated chapter/topic taxonomy: built a **vision-LLM pipeline** that parses PYQ documents end-to-end without manual tagging.

**Room-Scan Inventory Pipeline** | Swift, ARKit, Python, PyTorch, SAM2, YOLOv8-ORB | [Github](https://github.com)

- Solved a **small-object localization** problem at room scale: fused a LiDAR point cloud with SAM2 and a **fine-tuned YOLOv8-ORB** detector to pick out individual objects as small as a book spine from densely packed, low-texture shelves inside a full room reconstruction, then identified and priced each one through **multi-model LLM arbitration**.

## ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

- World's top 21 teams**, CANSAT'23 (NASA / AAS) | **India's top 12**, SIF Space Hackathon finalist (ISRO / IISF).
- 1st Position**, Blockchain Hackathon, Cognizance'24 IIT Roorkee | **Best Blockchain Project**, Frosthack'23 IIT Mandi | **1st Position**, Astrophysics Hackathon (Merope).
- Co-coordinator**, STAC IIT Mandi | **Media & Outreach Head**, Xpecto'23 | **Core Team**, E-Cell IIT Mandi.